

MINDS

RE CHANGING OUR LIVES

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TOP 10 INNOVATIONS OF THE LAST DECADE

Innovation is more than just invention; it involves commercialisation as well. IIM-Bangalore professor Rishikesh T. Krishnan lists his top picks.

BAJAJ PULSAR, 2001

It gave India's youth a blend of style, power and fuel economy by using technologies such as DTS-i and ExhaustEC

REVA, 2001

Using advanced energy management systems that optimise battery use, Reva leads the market in "clean" automobiles

ADVANCED LIGHT HELICOPTER, 2002

India's first major high-tech defence export from Hindustan Aeronautics Limited, it uses composite materials extensively

FUNGISOME, 2002

A liposomal anti-fungal drug developed by Lifecare Innovations, it has minimal effect on the kidneys

EDGE, 2002

The world's slimmest consumer watch by Titan, it is powered by a specially designed step motor and a chip to conserve battery power

PLASTIC BODY INVERTER, 2002

Designed by Su-kam, it uses advanced plastics to withstand high temperatures and yet gives a user-friendly finish

AVADIS, 2005

Designed by Strand Life Sciences, this data analysis software for biologists combines high-end visualisation and statistical tests

ETHERNET TRANSMISSION ON OPTICAL NETWORKING, 2005

Developed by Tejas Networks, it allows service providers to use networks created for voice to provide data services on telecom networks

MAHINDRA SHAAN, 2006

In-built trolley and new design features make this a unique cross-over vehicle usable across the year, giving farmers better utility

TATA NANO, 2009

Efficient use of space and redefining value engineering as well as design helped the Tatas create the world's best "value for money" car

tor only when his father apprenticed him to the workshop of the renowned Andrea del Verrocchio at the age of 15.

By and large, innovation has remained an individualistic enterprise, invariably flowering under personal care, such as that of Ashok Jhunjhunwala at IIT-Madras who heads the Telecommunication and Computer Networks Group (TeNet) or Anil Gupta at IIM-Ahmedabad who spearheaded the NRI. Innovative ideas are the enemies of restraints. So an M.S. Raju, an engineer from Visakhapatnam, can develop a camera mouse for the visually handicapped to read text messages and an Ankit Mehta from IIT-Bombay can work on an aerial device that security agencies are now planning to use. But as Gupta says, there aren't enough grassroots business incubator programmes in India. "Most of them are located at the top engineering and management schools, accessible only to a select few," he says. Ashok Khosla, chairman of Development Alternatives, a non-profit organisation that aims at creating large-scale sustainable livelihoods says, "There has always been a lack of government support, but to raise funds the innovators must also understand the market needs and invent products accordingly. If a product is not commercially viable, it's not an innovation but an invention."

How to break free of the restraint? Krishnan says there are three variables in any innovation—input, capacity and incentive. The incentives exist, as an innovator can gain monetarily, but input and capacity are still required. According to the Department of Science and Technology's 2007-08 statistics, investment in industrial R&D spending is dominated by two sectors, pharma (45 per cent) and transport (17 per cent). The rest is distributed over other sectors. This ratio has to become more equitable to encourage innovation in other sectors too. Only 26 per cent of the R&D spent is by the corporate sector, both public and private. In the private sector, a new breed of companies need to come up that are headed by young, technically qualified passionate people.

Only then can a Mansukhlal, from a small community of potters in Gujarat who has created refrigerators of clay, or Mohanlal, who improvised on marine diesel engines to make them lighter, flourish. As Peter Drucker said, "Technology is not about tools, it deals with how man works." And it's the innovator who makes the man's work easier.

with Natascha Shah and Sumaiya Khan